2017-2018 – DEEQA – TSE – Advanced Macroeconomics

Lecture 4 : Model Evaluation : Estimation and VARs

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Version 1.0
14/02/2018

Disclaimer

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1. Introduction : Model

- DSGE has the *structural form*

$$X_t = A_1 X_{t-1} + A_2 E_t[X_{t-1}] + A_3 Z_t$$
$$Z_t = \Gamma Z_{t-1} + \varepsilon_t$$

where X are endogenous variables, Z exogenous ones and ε *i.i.d.* unit-variance zero-mean shocks.
- Matrix coefficients are function of deep parameters Θ
- In general, expectations are not observed $\rightsquigarrow$ one needs a theory of expectations
- Often in the macro literature : Rational Expectations (RE)
- Then, if the model has a unique solution (we'll come back to that), it is of the form

$$X_t = AX_{t-1} + BZ_t$$
$$Z_t = \Gamma Z_{t-1} + \varepsilon_t$$

- This is the *reduced form*.
- A and B are functions of A_1, A_2, A_3 and Γ (*cross-equations restrictions*)

1. Introduction : Model

- In full generality, the model is written, with $Y = \begin{pmatrix} X \\ Z \end{pmatrix}$:

$$
\begin{aligned}
Y_t =& A_Y Y_{t-1} + B_y \varepsilon_t \\
\widetilde{Y}_t =& C_y Y_t + \nu_t
\end{aligned}
$$

- $\widetilde{Y}$ are the observables (might be of a lower dimension that Y) and ν is measurement error.
- For the moment, let's assume away $C_y \neq I$ and $\nu \neq 0$.
- We need to estimate $Y_t = A_Y Y_{t-1} + B_y \varepsilon_t$
- It is easy to estimate A_Y and $B_y \varepsilon_t$(OLS).
- What is difficult is to find the parameters Θ and the structural shocks ε.
- This is the *identification* problem.

1. Introduction : Methods for Validating Models

- Calibration
- Matching moments
  - GMM
  - SMM
  - Indirect Inference
- Full information methods
  - Maximum Likelihood
  - Bayesian methods
- Structural VARs

2. Calibration

- We have already given an example with the RBC model
  - × Use a set of empirical targets to pin down parameters
  - × Use a second set of targets to judge the model's empirical performances
  - × This is done without statistical foundations, and hence no hypothesis testing.
- One can do the some thing with statistical foundations $\rightsquigarrow$ matching moments

3. Matching Moments

- These are *limited information methods*
- Moment-matching procedures seek to find parameters θ that best enable the structural model to match a collection of preselected moments.
- Let $\widehat{\theta}$ be the estimated parameters.
- $\widehat{\theta}$ is a function of the data, therefore associated with uncertainty.
- One can therefore test that the selected moments can be plausibly interpreted as a random drawing from the underlying structural model.
- Various methods :
  - Generalized Method of Moments
  - Simulated Method of Moments
  - Indirect Inference

3. Matching Moments

Generalized Method of Moments, Hansen (1982)

- Data z_t, parameters θ of dimension $(q \times 1)$
- moments : real-values function $f(z_t, \theta)$, dimension r such that

$$E[f(z_t, \theta_0)] = 0 \qquad (\star)$$

- θ_0 is the true unknown value, f can be a mean, a correlation, etc...
- The idea of GMM is to choose $\widehat{\theta}$ such that the sample value of $(\star)$ is as close to zero as possible.
- That sample analog is

$$g(Z, \theta) = \frac{1}{T} \sum_{t=1}^{T} f(z_t, \theta)$$

  with $Z = [z'_T, z'_{T-1}, \ldots, z'_1]$
- Depending on $r \lesseqgtr q$, under-, just- or over-identification.

3. Matching Moments

Generalized Method of Moments, Hansen (1982)

- When $r \geq q$,
$$\min_{\theta} g(Z,\theta)' \times \Omega \times g(Z,\theta)$$
- Hansen (1982) : optimal Ω is the inverse of the variance-covariance matrix of sample mean $g(Z,\theta)$.

3. Matching Moments

Simulated Method of Moments

- Let y_t be models variables that correspond to empirical counterparts z_t
- let $h(z_t)$ be empirical targets
- Under GMM, $E[h(y_t, \theta)]$ can be computed analytically
- If it cannot be, one can simulate the model to derive samples of y_t and compute sample averages

$$h(Y, \theta) = \frac{1}{N} \sum_{t=1}^{N} h(y_t, \theta)$$

  with $Y = [z'_T, z'_{T-1}, \ldots, z'_1]$.
- This is an asymptotically consistent estimator of $E[h(y_t, \theta)]$
- The null hypothesis is that there exist a θ_0 such that

$$E[h(z_t)] = E[h(y_t, \theta_0)]$$

- To use the GMM notations, we have now

$$f(z_t, \theta_0) = h(z_t) - h(y_t, \theta_0)$$

and the sample analog of $g(Z, \theta_0)$ is

$$g(Z, \theta_0) = \left(\frac{1}{T} \sum_{t=1}^{T} h(z_t) \right) - \left(\frac{1}{N} \sum_{t=1}^{N} h(y_t, \theta_0) \right)$$

- plus some adjustments to make for Ω.

3. Matching Moments

Indirect Inference, Gourrieroux, Monfort, Renault (1993)

- Indirect Inference is moment matching, but the target moments are estimates from a reduced form model.
- For example :
  - The structural model has three variables and three shocks
  - One estimates a VAR with two variables and two shocks on the data and with simulated data
  - One finds the $\widehat{\theta}$ that minimises the distance between the data and model misspecified VAR

- Full information methods take the model as a complete statistical characterization of the data.
  - × They need distributional assumptions for the stochastic innovations. Inference is jointly conditional on the model and the distributional assumptions.
  - × Inference is done on the full range of empirical implications of the model, not only a subset
- One distinguishes classical approach (Maximum Likelihood) and Bayesian approach (which is also Likelihood analysis)

- Parameters are fixed but unknown
- Data are realisations of random drawing from the likelihood function
- Given a realisation of the data, parameters estimates max the likelihood function.
- Uncertainty in the parameters arises because the realized data are only one of many possibilities.
- The randomness in the data imparts randomness in the parameters estimates
- Main difficulty is how to numerically find the maximum (local maxima, flatness,...)

4. Full Information Methods

Bayesian approach

- Parameters are random and the data is fixed
- Estimation has for objective to make conditional probabilistic statements about the parameters, where the conditioning is with respect to the data.
- The probabilistic interpretation of the parameters allows to include a priori views about the parameters
  - We have a prior distribution about parameters
  - we have a likelihood function and the data
  - We can use BAYES' Rule to get a posterior distribution of the parameters.
  - We can look at the mean or mode of the posterior distribution to get parameters point estimates
- The question asked is how plausible is a given model relative to competing alternatives, given the observed data ?
- For classical approach, the question is how plausible are the observed data given the maintained null hypothesis (one model) ?
- Main challenge : computationally intensive

- A model is a simplification of reality. Strong test will almost always reject the model, except if one favors highly parametrized model.
- If we ignore the test, maximum likelihood is used to quantify the relative importance of different shocks.
- Then by estimation, we get estimates of the model parameters and variances of innovations which best fit the data.
- But if just identified, or close to just identified, then we don ?t learn if model is good : it is by assumption.

- Those previous methods are not always informative about the model fit
- If they fit badly (i.e. overidentification is rejected), often don ?t know why.
  - Does the model work well on some dimension and not on others ?
  - Maybe doesn ?t fit well because we forgot to include a shock ?
- Even if model fits well, it could be for the wrong reasons. By looking at data with all shocks considered together, may confounding effects (especially with limited information approach)
- This motivates looking at more economically informative methods : Structural VARs

- Here I illustrate the logic of shocks identification is structural VARs.
- As the model used for identification, we choose the AD/AS model of the Neoclassical synthesis, that was the standard workhorse of macroecomics until the mid 70's, and that is still the common wisdom among many politicians and journalists.
- IS-LM and AD-AS models
- General Equilibrium without explicit micro-foundations of individual decisions and description of markets organization

- AD : Level of aggregate demand as a function of the general price level ($M =$ money supply, G=govt expenditures)

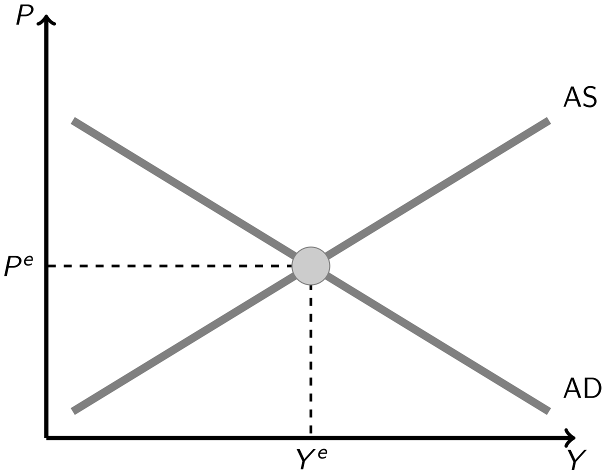
$$P = a_0 - a_1 Y + a_2 M + a_3 G \tag{1}$$

- AS : Level of aggregate supply as a function of the general price level ($Q =$ productivity)

$$Y = b_0 + b_1 P + b_2 Q \tag{2}$$

- Note : Y = value added = income = expenditures (regarding only final goods)

FIGURE 1 – Equilibrium of the AD-AS Model



- One can compute the solution and multipliers

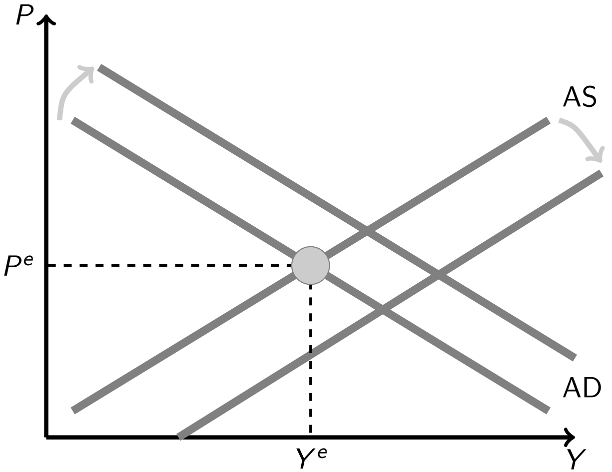
$$
\begin{aligned}
P &= \left(\frac{a_0 - a_1 b_0}{1 + a_1 b_1}\right) + \left(\frac{a_2}{1 + a_1 b_1}\right) M + \left(\frac{a_3}{1 + a_1 b_1}\right) G - \left(\frac{a_1 b_2}{1 + a_1 b_1}\right) Q \\
Y &= \left(\frac{b_0 + b_1 a_0}{1 + a_1 b_1}\right) + \left(\frac{b_1 a_2}{1 + a_1 b_1}\right) M + \left(\frac{b_1 a_3}{1 + a_1 b_1}\right) G + \left(\frac{b_2}{1 + a_1 b_1}\right) Q
\end{aligned}
$$

- The model is mainly used for policy evaluation : $\frac{\partial Y}{\partial G}$, $\frac{\partial Y}{\partial M}$, $\frac{\partial Y}{\partial Q}$, $\frac{\partial P}{\partial M}$, *etc...* (oil price shock, monetary expansion, ...)
- Most of the debate was then the size and signs of the multipliers.
- This model have foundations, although not micro-foundations : IS-LM for AD and the functioning of the labor market for AS.

5. SVARs : an AD-AS example

A First Look at the AD-AS Model

FIGURE 2 – Shocks and Policies in the AD-AS Model



- Let's use a Blanchard & Quah (AER 1989) (BQ) type of analysis to evaluate the relative importance of "supply" and "demand" shock
- The idea is to decompose any movement of the economy as the consequence of 2 orthogonal shocks : a demand shock and a supply one.

- Assume that the model economy is the following AD-AS :

$$\begin{cases} P = -\alpha Y + \varepsilon^D & (AD) \\ P = \beta Y - \varepsilon^S & (AS) \end{cases}$$

- α and β are positive constants
- Shocks are zero-mean stochastic variables

- Remark : How do we obtain such a linear model in which the non-stochastic solution is $Y = P = 0$?

$$\begin{cases} \mathcal{P} & = & A \;\; \mathcal{Y}^{-\alpha} \;\; e^{\varepsilon^D} & \qquad\qquad (AD) \\ \mathcal{P} & = & B \;\; \mathcal{Y}^{\beta} \;\; e^{-\varepsilon^S} & \qquad\qquad (AS) \end{cases}$$

- Compute non-stochastic solution :

$$\begin{cases} \overline{\mathcal{Y}} & = & \left(\frac{A}{B}\right)^{\frac{1}{\alpha+\beta}} \\ \overline{\mathcal{P}} & = & A^{\frac{\beta}{\alpha+\beta}} B^{\frac{\alpha}{\alpha+\beta}} \end{cases}$$

- Denote $X = \log \mathcal{X} - \log \overline{\mathcal{X}}$
- We then obtain

$$\begin{cases} P &= -\alpha Y + \varepsilon^D \qquad (AD) \\ P &= \beta Y - \varepsilon^S \qquad (AS) \end{cases}$$

- Assume that we know α and β.
- Then, one can identify demand and supply shocks (namely ε^D and ε^S)

FIGURE 3 – Observation : The economy went from A to B and C

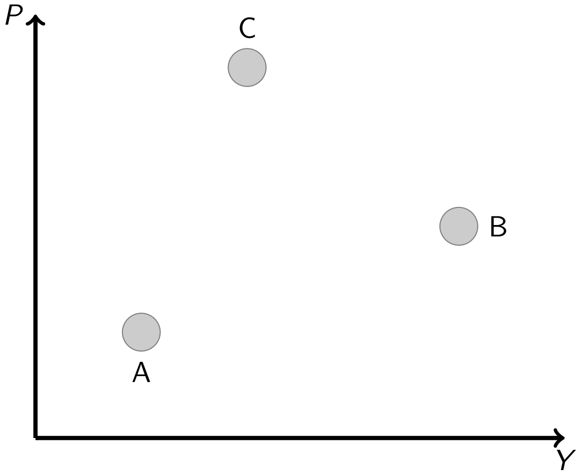


FIGURE 4 – We aim at putting names (stories) on those wiggling arrows

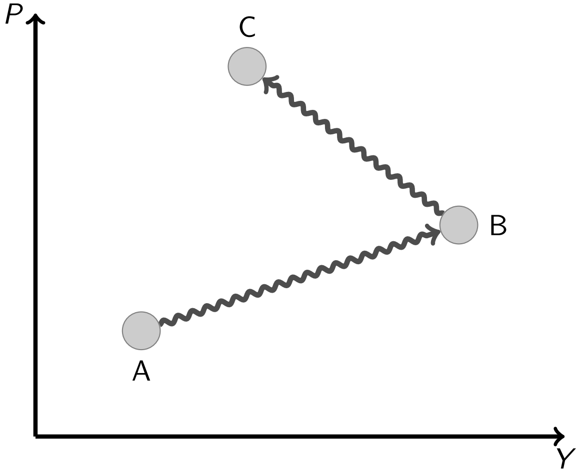


FIGURE 5 – The AD-AS model provides us with a theory of economic fluctuations (the wiggling arrows) with the help of the gray shifters

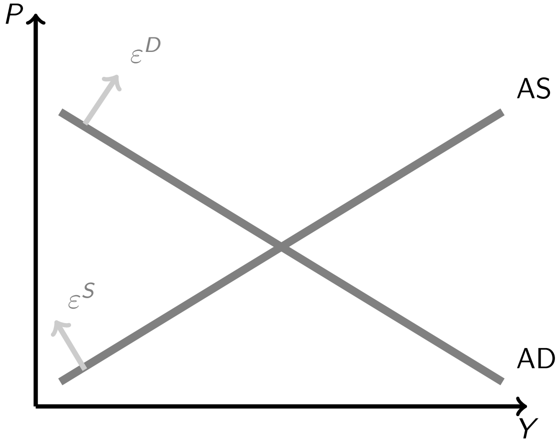


FIGURE 6 – Each Observation is at the crossing of one AD and one AS curve

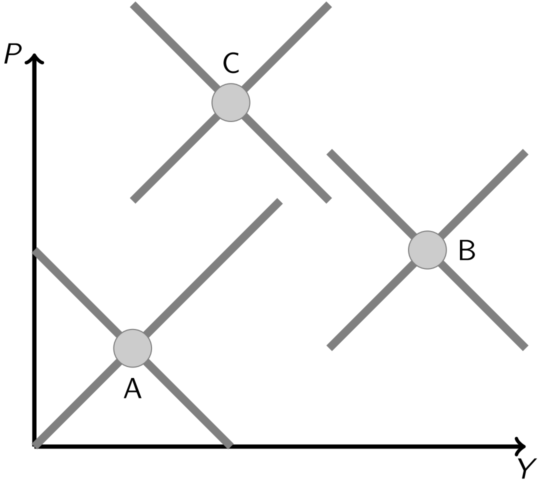


FIGURE 7 – This is the structural interpretation of the move from A to B

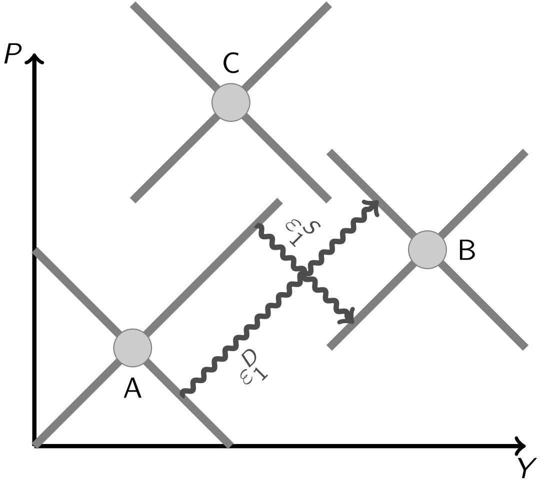


FIGURE 8 – This is the structural interpretation of the move from B to C

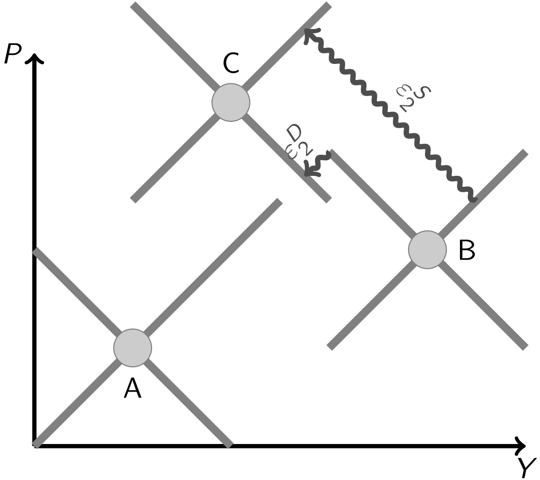
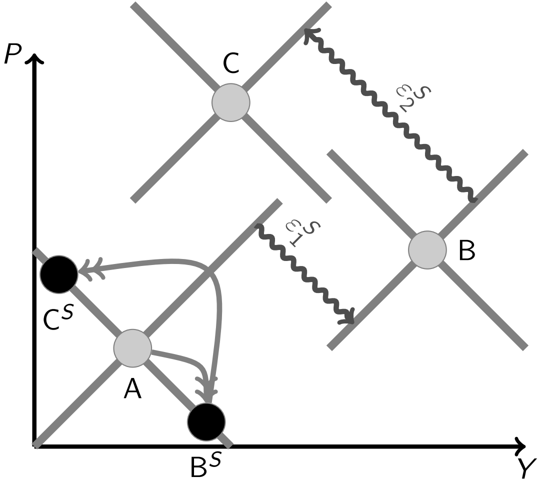


FIGURE 9 – Counterfactual : What would have happen absent of demand shocks



- The algebra is even more simple. If one solves the model

$$\begin{cases} P = -\alpha Y + \varepsilon^D & (AD) \\ P = \beta Y - \varepsilon^S & (AS) \end{cases}$$

one gets

$$\begin{cases} P = \frac{\beta}{\alpha+\beta}\varepsilon^D - \frac{\alpha}{\alpha+\beta}\varepsilon^S \\ Y = \frac{1}{\alpha+\beta}\varepsilon^D + \frac{1}{\alpha+\beta}\varepsilon^S \end{cases}$$

- When one observes Y and P, this is a set of 2 equations with 2 unknowns, ε^D and ε^S $\rightsquigarrow$ one can recover the structural shocks.
- The problem is that in the real world, we do not know α and β

- One way could be to estimate each of the two equations using instrumental variables (oil price when estimating AD, money supply or Gvt expenditures when estimating AS)
- But it is very unlikely that this very simple and static model captures a significant part of the economy variance

- Assume that the economy is best described by the following dynamic model :

$$\left\{\begin{array}{lll} P_t & = & \alpha_0^D Y_t + \alpha_1^D Y_{t-1} + \alpha_2^D Y_{t-2} + \cdots + \alpha_N^D Y_{t-N} \\ & + & \beta_1^D P_{t-1} + \beta_2^D P_{t-2} + \cdots + \beta_N^D P_{t-N} + \varepsilon_t^D \qquad (AD) \\ P_t & = & \alpha_0^S Y_t + \alpha_1^S Y_{t-1} + \alpha_2^S Y_{t-2} + \cdots + \alpha_N^S Y_{t-N} \\ & + & \beta_1^S P_{t-1} + \beta_2^S P_{t-2} + \cdots + \beta_N^S P_{t-N} + \varepsilon_t^S \qquad (AS) \end{array}\right.$$

- Demand and Supply shocks are independent.

- Consider the simple dynamic model
$$X_t = \rho X_{t-1} + \varepsilon_t \tag{1}$$
with $0 < \rho < 1$ and ε *iid*, with $E(\varepsilon_t) = 0$, $V(\varepsilon_t) = \sigma^2$
- (1) is the autoregressive (AR) representation of the model.
- One can write a moving average (MA) representation, which expresses X_t as a function of the current and past values of ε.

$$\begin{aligned} X_t &= \rho X_{t-1} + \varepsilon_t \\ &= \rho(\rho X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \\ &= \rho^2 X_{t-2} + \rho\varepsilon_{t-1} + \varepsilon_t \\ &= \rho^3 X_{t-3} + \rho^2\varepsilon_{t-2} + \rho\varepsilon_{t-1} + \varepsilon_t \\ &= \rho^n X_{t-n} + \rho^{n-1}\varepsilon_{t-n+1} + \cdots + \rho\varepsilon_{t-1} + \varepsilon_t \end{aligned}$$

- When $n \to \infty$, as $0 < \rho < 1$
$$X_t = \sum_{i=0}^{\infty} \rho^i \varepsilon_{t-i}$$

- Making use of the lag operator notation : $LX_t = X_{t-1}$, $L^i X_t = X_{t-i}$, $i \in \mathcal{Z}$

$$
\begin{aligned}
X_t &= \rho X_{t-1} + \varepsilon_t \\
X_t &= \rho L X_t + \varepsilon_t \\
(1-\rho L) X_t &= \varepsilon_t \\
X_t &= \tfrac{1}{1-\rho L} \varepsilon_t \\
X_t &= (1 + \rho L + \rho^2 L^2 + \cdots + \rho^n L^n + \cdots) \varepsilon_t \\
X_t &= \varepsilon_t + \rho \varepsilon_{t-1} + \cdots + \rho^{n-1} \varepsilon_{t-n+1} + \rho^n \varepsilon_{t-n} + \cdots
\end{aligned}
$$

- More generally, a univariate MA(∞) is denoted :

$$X_t = a_0 \varepsilon_t + a_1 \varepsilon_{t-1} + \cdots + a_{n-1} \varepsilon_{t-n+1} + a_n \varepsilon_{t-n} + \cdots$$

- One can use the MA representation of the model to compute various indicators.

  Impulse response function : what is the dynamic effect of a shock ? Assume that for $t = -\infty$ to 0, $\varepsilon_t = 0$, such that $X_0 = 0$.
- The shock is $\varepsilon_1 = 1$, with $\varepsilon_t = 0$ for $t > 1$.
- Then $X_1 = 1$, $X_2 = \rho$, $X_3 = \rho^2$, etc...
- Note that the IRF is given by the coefficients of the MA representation.

Forecast Error Variance : The mathematical expectation of X_{t+1} based on the information of period t is denoted $E_t(X_{t+1})$.

$$\begin{aligned} E_t(X_{t+1}) &= E_t(\rho X_t + \varepsilon_{t+1}) \\ &= \rho \underbrace{E_t X_t}_{X_t} + \underbrace{E_t \varepsilon_{t+1}}_{0} \end{aligned}$$

- The one-step forecast error if X_t is $FE_1 = X_{t+1} - E_t(X_{t+1}) = \varepsilon_{t+1}$.
- $E_t(FE_1) = 0$ and $V(FE_1) = \sigma^2$
- Similarly, $FE_k = X_{t+k} - E_t(X_{t+k}) = \rho^{k-1}\varepsilon_{t+1} + \rho^{k-2}\varepsilon_{t+2} + \cdots + \varepsilon_{t+k}$, $E_t(FE_k) = 0$ and $V(FE_k) = (\rho^{2(k-1)} + \rho^{2(k-2)} + \cdots + \rho^2 + 1)\sigma^2$
- Note again that these statistics are functions of the MA coefficients.
- For example, $V(FE_k) = (a_k^2 + a_{k-1}^2 + \cdots + a_1^2 + a_0^2)\sigma^2$

- Assume that the economy is best described by the following dynamic model :

$$\begin{cases} P_t & = & \alpha_0^D Y_t + \alpha_1^D Y_{t-1} + \alpha_2^D Y_{t-2} + \cdots + \alpha_N^D Y_{t-N} \\ & + & \beta_1^D P_{t-1} + \beta_2^D P_{t-2} + \cdots + \beta_N^D P_{t-N} + \varepsilon_t^D \qquad (AD) \\ P_t & = & \alpha_0^S Y_t + \alpha_1^S Y_{t-1} + \alpha_2^S Y_{t-2} + \cdots + \alpha_N^S Y_{t-N} \\ & + & \beta_1^S P_{t-1} + \beta_2^S P_{t-2} + \cdots + \beta_S^P P_{t-N} + \varepsilon_t^S \qquad (AS) \end{cases}$$

- making use of the lag operator notation :
  $LX_t = X_{t-1}$, $L^i X_t = X_{t-i}$, $i \ \in \mathcal{Z}$

- and rearranging terms

$$\begin{cases} Y_t &= (\alpha_1^Y + \alpha_2^Y L + \cdots + \alpha_N^Y L^{N-1}) Y_{t-1} \\ & + (\beta_1^Y + \beta_2^Y L + \cdots + \beta_N^Y L^{N-1}) P_{t-1} + \gamma_D^Y \varepsilon_t^D + \gamma_S^Y \varepsilon_t^S \qquad (1) \\ P_t &= (\alpha_1^P + \alpha_2^P L + \cdots + \alpha_N^P L^{N-1}) Y_{t-1} \\ & + (\beta_1^P + \beta_2^P L + \cdots + \beta_N^P L^{N-1}) P_{t-1} + \gamma_D^P \varepsilon_t^D + \gamma_S^P \varepsilon_t^S \qquad (2) \end{cases}$$

- or equivalently

$$X_t = \widehat{A}(L) X_{t-1} + B\varepsilon_t$$

  with $X_t = (Y_t, P_t)'$ and $\varepsilon_t = (\varepsilon_t^D, \varepsilon_t^S)$
- This is the VAR (Vector AutoRegressive) representation of the equilibrium.

- It is convenient to work with the VMA (Vectorial Moving Average) representation

$$X_t = \frac{B}{I - \widehat{A}(L)L}\varepsilon_t$$

or

$$X(t) = \sum_{j=0}^{\infty} A(j)\varepsilon_{t-j}$$

with $\text{Var}(\varepsilon_t) = I$ and

$$A(j) = \begin{pmatrix} a_{11}(j) & a_{12}(j) \\ a_{21}(j) & a_{22}(j) \end{pmatrix}$$

- This dynamic model, if derived from a structural one, puts a lot of restrictions on the sequence of $A(\cdot)$

- Here I derive some summary statistics from the VMA representation
- Let us consider output. We have

$$Y_t = \sum_{j=0}^{\infty} a_{11}(j)\varepsilon_{t-j}^{D} + \sum_{j=0}^{\infty} a_{12}(j)\varepsilon_{t-j}^{S}$$

- The IRF to a demand shock is $\{a_{11}(0), a_{11}(1), a_{11}(2), ...\}$ and the IRF to a supply shock is $\{a_{12}(0), a_{12}(1), a_{12}(2), ...\}$

- The Forecast Error in predicting Y at horizon 1 is

$$Y_{t+1} - E_t Y_{t+1} = a_{11}(0)\varepsilon^D_{t+1} + a_{12}(0)\varepsilon^S_{t+1}$$

and the share of the variance of FE at horizon 1 attributable to the demand shock is

$$\frac{a^2_{11}(0)}{a^2_{11}(0) + a^2_{12}(0)}$$

(the variances of the structural shocks is normalized to 1).

- At horizon k, this share is

$$\frac{\sum_{j=0}^{k} a^2_{11}(j)}{\sum_{j=0}^{k} a^2_{11}(k) + \sum_{j=0}^{k} a^2_{12}(j)}$$

- Historical decomposition : what would have happen if only demand or supply shocks have been there ?

$$Y_t^D = \sum_{j=0}^{\infty} a_{11}(j)\varepsilon_{t-j}^D$$

$$Y_t^S = \sum_{j=0}^{\infty} a_{12}(j)\varepsilon_{t-j}^S$$

- Let us estimate a VAR model with Y and P.

$$X_t = \widetilde{A}(L)X_{t-1} + \nu_t$$

with $\mathrm{Var}(\nu) = \Omega$ and $C(0) = I$ by normalization.

- Note that the νs are different from the ϵs (they are an unknown linear combination of the ϵs)
- From this estimated VAR form, one can recover the following *non structural* (or *reduced form)* VMA representation

$$X(t) = \sum_{j=0}^{\infty} C(j)\nu_{t-j}$$

- How can ν be cut into two orthogonal pieces that we will label demand and supply shocks ?

- Compare this VMA representation with the *structural* one

$$X(t) = \sum_{j=0}^{\infty} A(j)\varepsilon_{t-j}$$

- As the two equations are representations of the same model,

$$\nu = A(0)\varepsilon \text{ and } A(j) = C(j)A(0) \text{ for } j > 0.$$

- Estimation gives us C.
- Once we know $A(0)$, we have everything. We have therefore 4 unknowns : $a_{11}(0), a_{12}(0)$, $a_{21}(0)$ and $a_{22}(0)$.

- How do we get $A(0)$? First, if $\nu = A(0)\varepsilon$, then ν and $A(0)\varepsilon$ have the same variance-covariance matrix.
- The one of ν is the Ω (estimated). The one of ε is I by assumption.
- Therefore, one has

$$V(A(0)\varepsilon) = V(\nu) \Longleftrightarrow A(0)A(0)' = \Omega$$

or

$$\begin{pmatrix} a_{11}(0) & a_{12}(0) \\ a_{21}(0) & a_{22}(0) \end{pmatrix} \times \begin{pmatrix} a_{11}(0) & a_{12}(0) \\ a_{21}(0) & a_{22}(0) \end{pmatrix}' = \begin{pmatrix} \omega_{11}(0) & \omega_{12}(0) \\ \omega_{12}(0) & \omega_{22}(0) \end{pmatrix}$$

- This gives us 3 equations (because Ω and $A(0)A(0)'$ are symmetrical) for 4 unknowns (the 4 coefficients of $A(0)$)

- We need one identifying assumption, that will allow us to separate aggregate demand shocks from aggregate supply ones.
- This last condition cannot come from the math. It has to be a restriction imposed by the economist, based on some "reasonable" property of the economy.

- Here only one extra restriction is needed because we have a 2-variables VAR. It could be more in larger models.
- This restriction should come from a model.
- Blanchard & Quah (1989) proposed the following restriction : *Only supply shocks affect output in the long run* or in other words *Demand shocks do not affect output in the long run*.
- The long run effect of a demand shock is $a_{11}(\infty)$
- But $A(\infty) = C(\infty)A(0)$ or

$$\begin{pmatrix} a_{11}(\infty) & a_{12}(\infty) \\ a_{21}(\infty) & a_{22}(\infty) \end{pmatrix} = \begin{pmatrix} c_{11}(\infty) & c_{12}(\infty) \\ c_{21}(\infty) & c_{22}(\infty) \end{pmatrix} \times \begin{pmatrix} a_{11}(0) & a_{12}(0) \\ a_{21}(0) & a_{22}(0) \end{pmatrix}$$

- The fourth restriction is therefore

$$c_{11}(\infty)a_{11}(0) + c_{21}(\infty)a_{21}(0) = 0$$

- Recall that the $c_{ij}(\infty)$ are known (from estimation).
- We can therefore compute $A(0)$.
- Once we have $A(0)$, and the estimated VAR, we can compute IRF to shocks and Forecast Error Variance decomposition

- Here, I have assumed that both the model and the estimated VAR could be written and estimated as

$$X_t = \widehat{A}(L)X_{t-1} + B\varepsilon_t$$

- Both theory and estimation could imply that there are some constants or trends in this expression

$$X_t = \widehat{A}(L)X_{t-1} + B\varepsilon_t + K_1 + K_2 t$$

- There is also an issue about the best way to specify the VAR if the series are non stationary (ie if shocks have permanent effect, which is the case here). It might be more efficient to specify the VAR in difference

$$(1 - L)X_t = (1 - L)\widehat{A}(L)X_{t-1} + B\varepsilon_t + K_1$$

- In each of these case, although the algebra is more tedious, the same results than the one I have presented here apply.

- Data : US 1947Q1-2015Q4 quarterly data
- Output is Real GDP per capita, Prices series is the GNP deflator.
- With some abuse of the interpretation of the AD-AS model, we consider not P and Y but ΔP and ΔY.
- Take 12 lags in the VAR

5. SVARs : an AD-AS example

Data

Figure 10 – US Output and Prices, 1947Q1-2015Q4

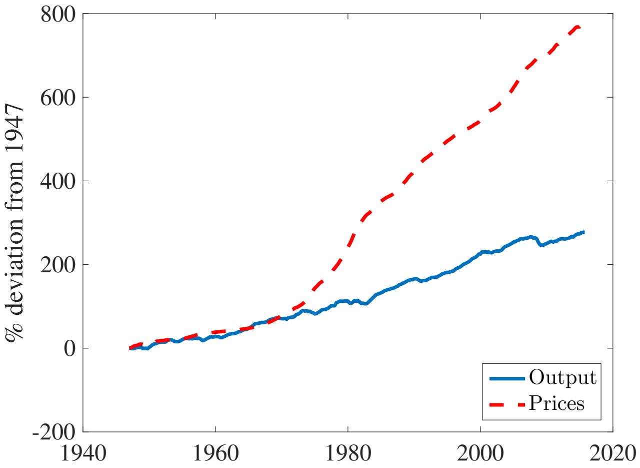
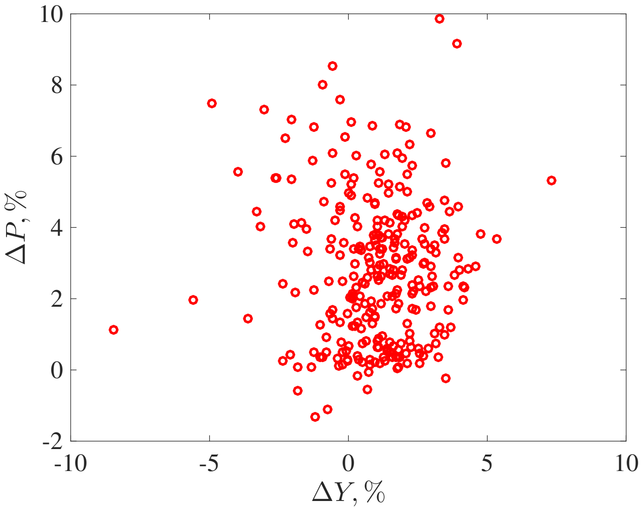
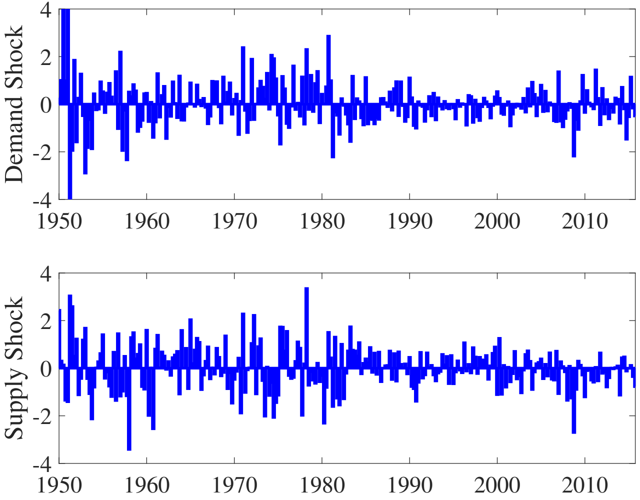


FIGURE 11 – US Growth Rates of Output and Prices, 1947Q1-2015Q4



Results : IRF and Variance Decomposition

FIGURE 12 – Estimated Shocks



5. SVARs : an AD-AS example

Results : IRF and Variance Decomposition

FIGURE 13 – IRF

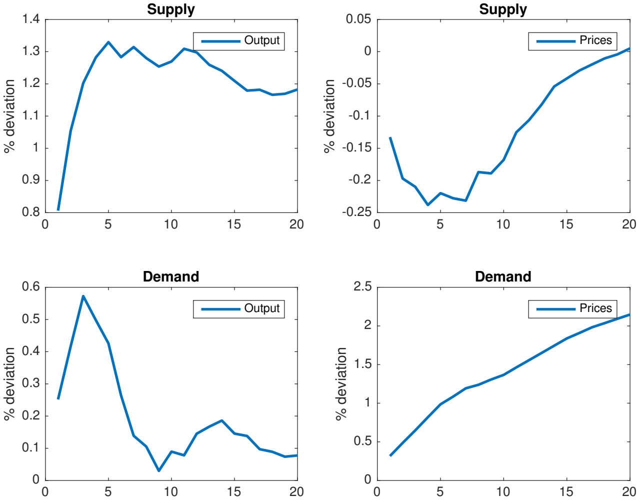


Figure 14 – Output FE Variance Decomposition

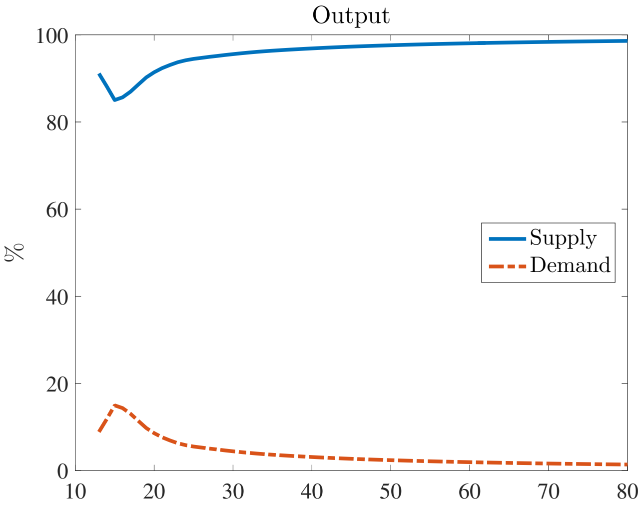
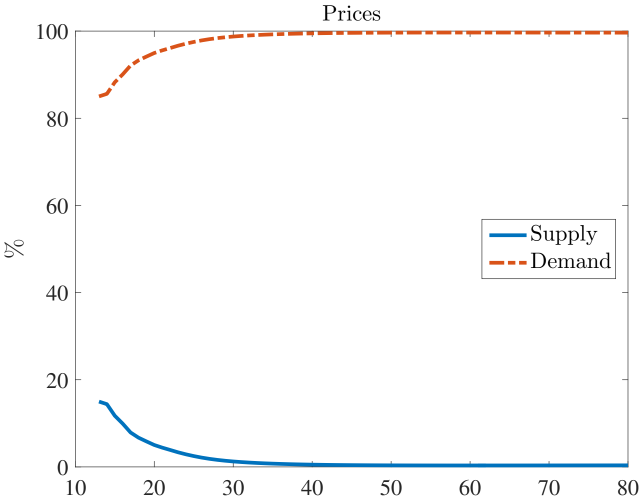


FIGURE 15 – Prices FE Variance Decomposition

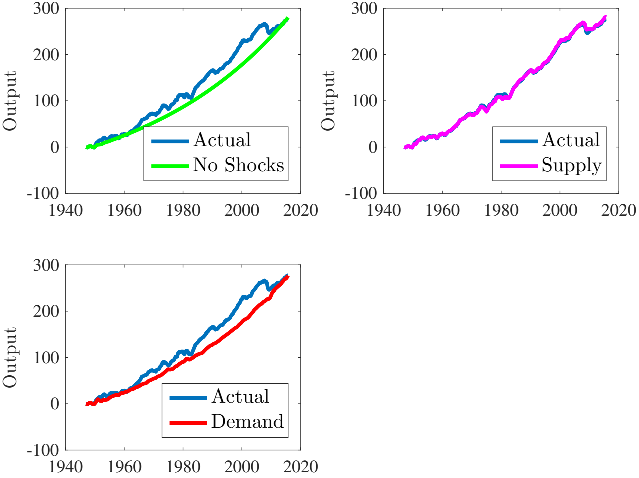


- We use the following color code :
  - **Blue : actual series**
  - **Green : the series with no shocks**
  - **Pink : the series with only the supply shocks**
  - **Red : the series with only the demand shocks**

5. SVARs : an AD-AS example

Results : Historical Decomposition

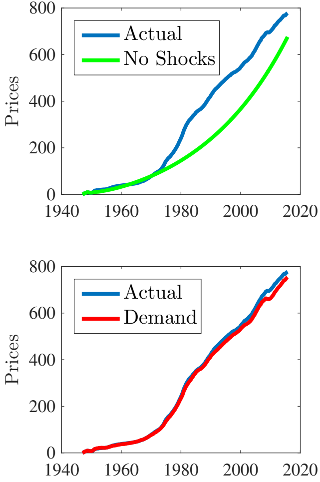
FIGURE 16 – Whole Sample



5. SVARs : an AD-AS example

Results : Historical Decomposition

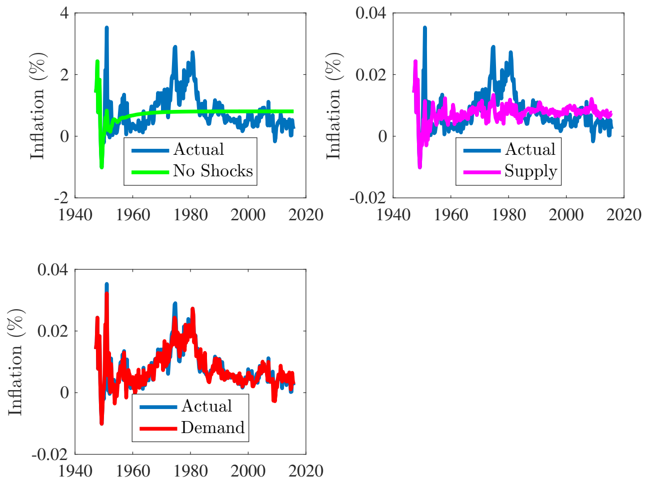
FIGURE 17 – Whole Sample



5. SVARs : an AD-AS example

Results : Historical Decomposition

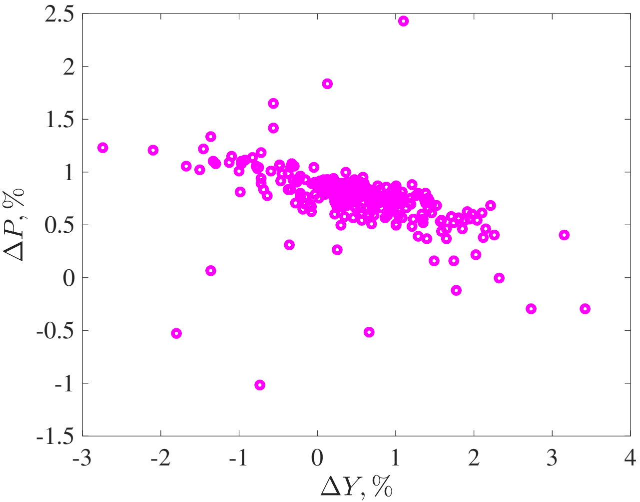
FIGURE 18 – Whole Sample



5. SVARs : an AD-AS example

Results : Historical Decomposition

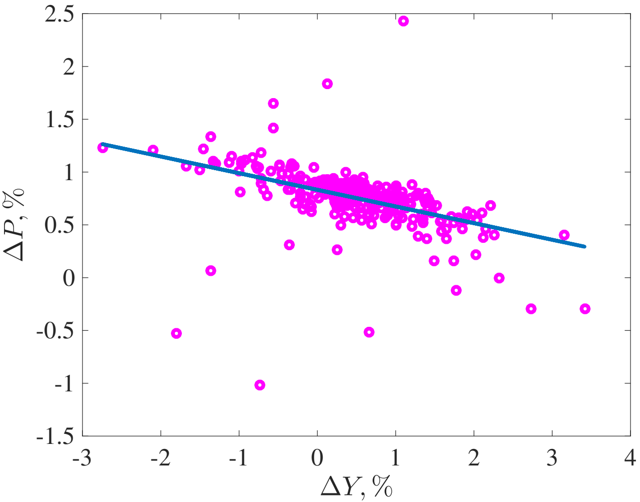
FIGURE 19 – Whole Sample - Supply Shocks Only



5. SVARs : an AD-AS example

Results : Historical Decomposition

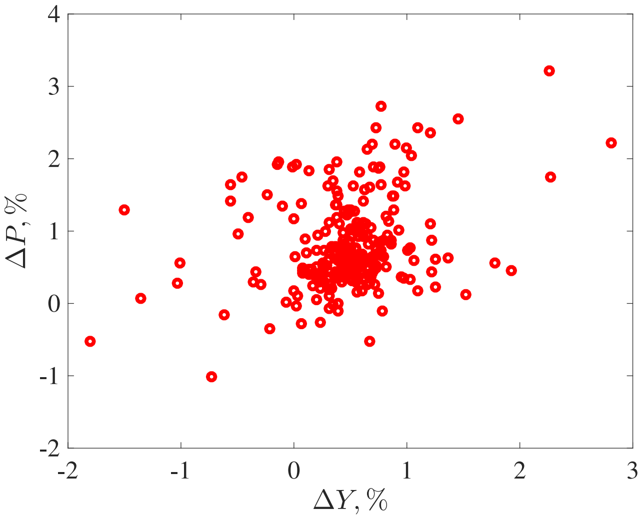
FIGURE 20 – Whole Sample - Supply Shocks Only



5. SVARs : an AD-AS example

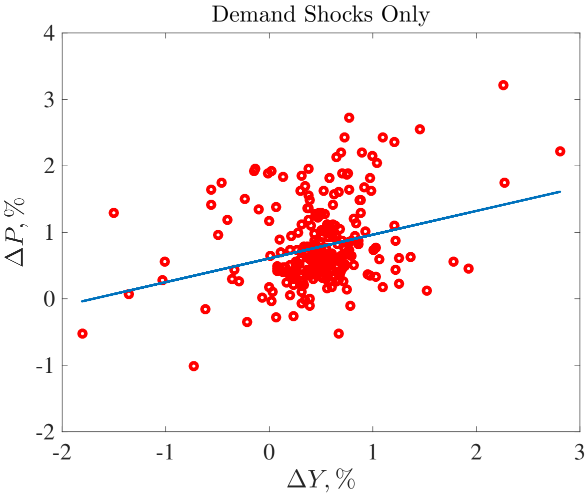
Results : Historical Decomposition

FIGURE 21 – Whole Sample - Demand Shocks Only



Results : Historical Decomposition

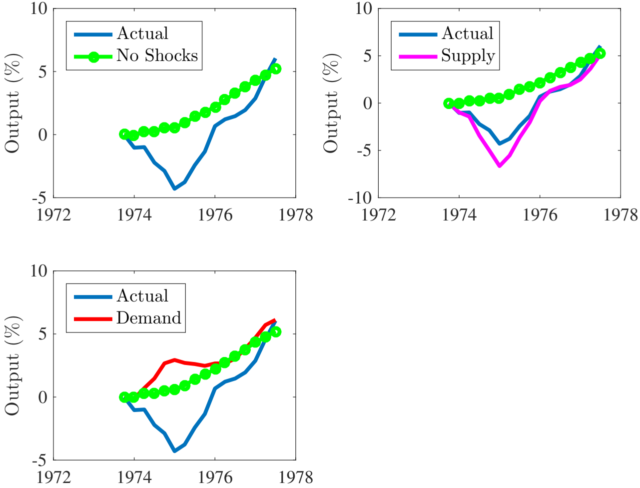
FIGURE 22 – Whole Sample - Demand Shocks Only



5. SVARs : an AD-AS example

Results : Historical Decomposition

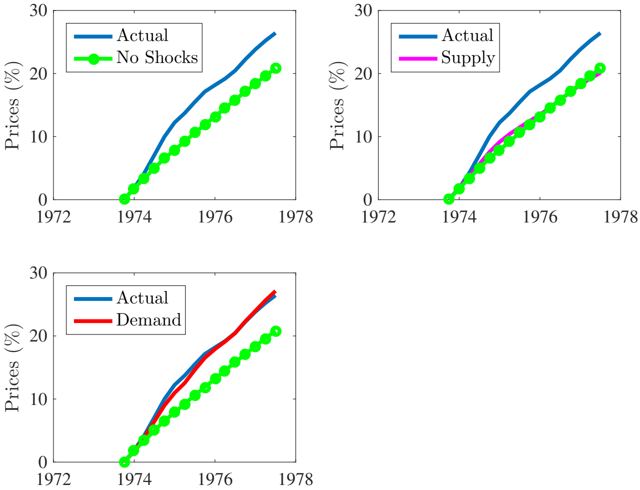
FIGURE 23 – First Oil Shock



5. SVARs : an AD-AS example

Results : Historical Decomposition

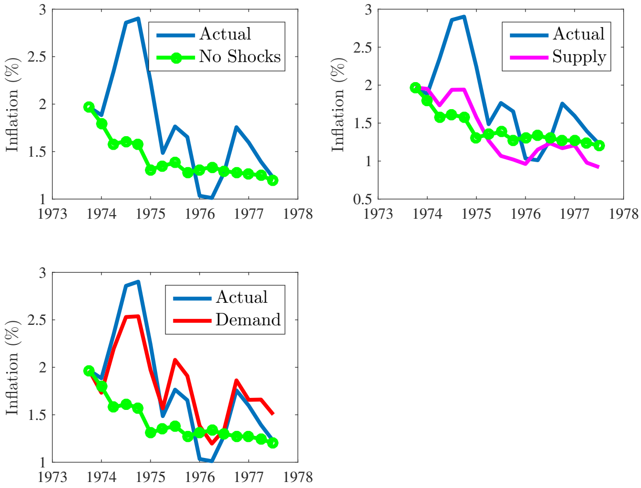
FIGURE 24 – First Oil Shock



5. SVARs : an AD-AS example

Results : Historical Decomposition

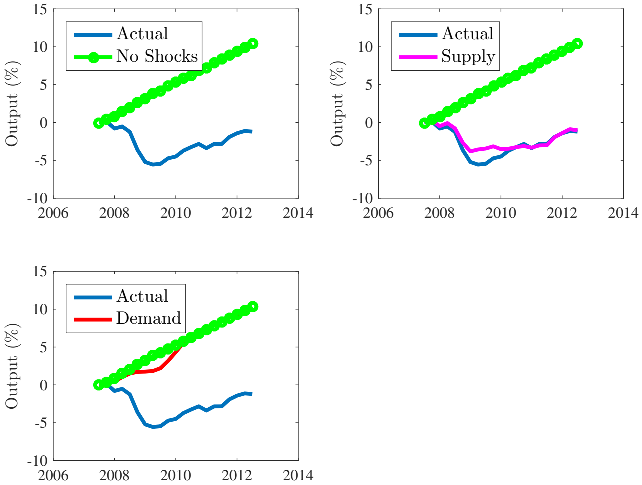
FIGURE 25 – First Oil Shock



5. SVARs : an AD-AS example

Results : Historical Decomposition

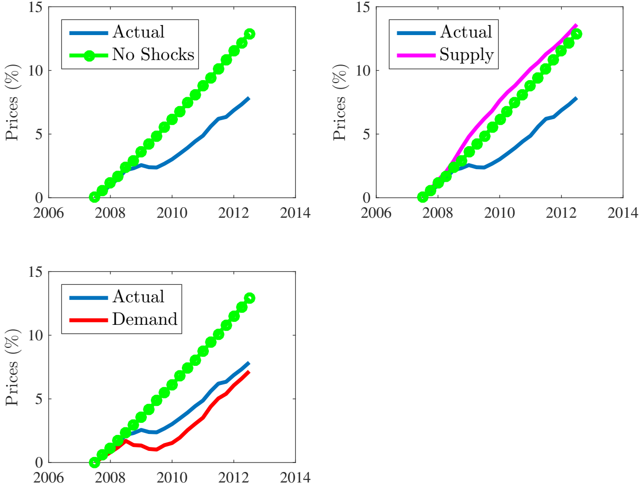
FIGURE 26 – The Great Recession



5. SVARs : an AD-AS example

Results : Historical Decomposition

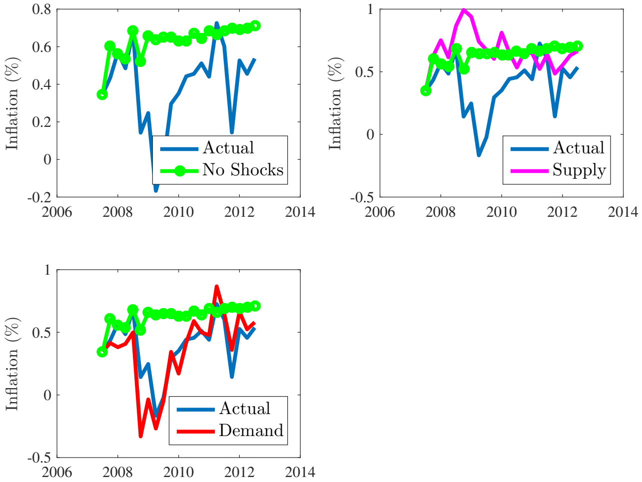
FIGURE 27 – The Great Recession



5. SVARs : an AD-AS example

Results : Historical Decomposition

Figure 28 – The Great Recession



- A DSGE model has a solution that can be written as a Structural Moving Average

$$Y_t = \Gamma(L)v_t$$

where the variance of v is the identity matrix and $\Gamma(L) = \Gamma_0 + \Gamma_1 L + \Gamma_2 L^2 + \cdots$

- When one estimates a VAR on the data, one obtains the Wold representation

$$Y_t = C(L)\varepsilon_t$$

were $C(L) = I + C_1 L + C_2 L^2 + \cdots$ and where the variance of ε is Σ_ε

- The problem of identification in SVARs is to recover $\Gamma(L)$ from $C(L)$ and Σ_ε.
- It is indeed a quite simple non linear set of equations to solve, once we impose some restrictions

$$Y_t = \Gamma(L)v_t$$
$$Y_t = C(L)\varepsilon_t$$

- Rewrite the SMA as
$$Y_t = \Gamma(L)\Gamma_0^{-1}\Gamma_0 v_t$$
- That representation is equivalent to the Wold one iif
$$\varepsilon_t = \Gamma_0 v_t$$
$$C(L) = \Gamma(L)\Gamma_0^{-1}$$
$$\Gamma_0\Gamma_0' = \Sigma_\varepsilon$$

$$Y_t = \Gamma(L)v_t$$
$$Y_t = C(L)\varepsilon_t$$

- we need to solve for the coefficients of Γ_0 such that

$$C(L) = \Gamma(L)\Gamma_0^{-1}$$
$$\Gamma_0\Gamma_0' = \Sigma_\varepsilon$$

- Because Σ_ε is symmetric, this gives us $\frac{n(n-1)}{2}$ too many parameters $\rightsquigarrow$ we need at least $\frac{n(n-1)}{2}$ restrictions if we want to identify all the shocks.

$$Y_t = \Gamma(L)v_t \qquad\qquad Y_t = C(L)\varepsilon_t$$

- We can do various type of restrictions :
  - × Short run (directly on Γ_0) as in "*the monetary shock has no impact effect on ouptut*"
  - × Long run (directly on Γ_∞) as in "*the monetary shock has no long effect on output*"
  - × Restrictions need not to be zero restrictions, as in "*prices and ages respond by the same amount to a demand shock on impact or in the long run*"
  - × Restrictions need not to be either on impact or in the long run, as in "*the trade balance reaches zero in 3 quarters following a trade shock*"
  - × Restrictions need not te recover all the shocks, as in "*the demand shock is the one that is orthogonal to all the others and explains the maximum of the variance of hours worked at a 20 quarters horizon*"
  - × Restrictions can be qualitative (set identification) as in "*the monetary shock is the only shock that moves output and prices up and the nominal interest down*"
- One can also mix those restrictions
- It is **bad language** to present restriction as variables ordering

- VAR specification (order, cointegration, VECM, estimation in levels,...) $\rightsquigarrow$ this is econometrics.
- Fundamental representation and invertibilty (see later)